

Week 2 Project: Advanced Line Calculations

Thermal Analysis, Voltage Drop, and Conductor Sizing

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Contents

1	Introduction	3
2	Thermal Equilibrium and Heat Transfer Analysis	3
2.1	The Fundamental Heat Balance Equation	3
2.2	Heat Generation ($P_{\text{generated}}$)	4
2.3	Heat Dissipation ($P_{\text{dissipated}}$)	4
2.3.1	Underground Installation	5
2.3.2	Overhead Installation	5
2.4	AC Resistance and Maximum Current	8
2.5	Thermal Analysis Results	8
3	Advanced Voltage Drop Analysis	8
3.1	Blondel's Formula	8
3.2	REBT Compliance and Voltage Profile	9
4	Conductor Sizing and Economic Analysis	11
4.1	Technical and Economic Sizing Criteria	11
4.2	Lifecycle Cost Analysis	11
4.3	Sizing Analysis Results	12
5	Conclusion	12
A	References	13
B	Tables and Assumed Values	14
B.1	List of Tables	14
	List of Tables	14
B.2	Summary of Material Properties	14
C	Formulas	15
C.1	AC Resistance at Temperature	15
C.2	Blondel's Formula (Three-Phase)	15
C.3	Required Section (Voltage Drop)	15
C.4	Lifecycle Cost	15
D	MATLAB Source Code	16
MATLAB Source Code		16
D.1	week2_master.m	16
D.2	E0_Run_Analysis_and_Report.m	17
D.3	E1_ConductorHeatingAnalysis.m	18
D.4	E2_MaximumCurrent.m	19
D.5	E3_ResistanceTemperatureCorrection.m	20
D.6	E3b_AC_Resistance_Factor.m	20
D.7	E4_ThermalEquilibrium.m	20
D.8	E5_HeatingCurves.m	21
D.9	E6_TransientHeating.m	21
D.10	E7_HeatDissipation.m	21

D.11 E8_GroupingFactor.m	22
D.12 F1_VoltageDropAnalysis.m	23
D.13 F2_calculateExactVoltageDrop.m	23
D.14 F3_calculateSimplifiedVoltageDrop.m	23
D.15 F4_checkREBTCompliance.m	23
D.16 F4b_checkInstallationCompliance.m	24
D.17 F5_plotVoltageProfile.m	24
D.18 F6_plotInstallationProfile.m	24
D.19 G1_ConductorSizingAnalysis.m	25
D.20 G2_calculateRequiredSection.m	25
D.21 G3_selectStandardSection.m	26
D.22 G4_calculateLifecycleCost.m	26
D.23 G5_plotEconomicAnalysis.m	26
D.24 G6_verifyFinalDesign.m	27

1 Introduction

This report details the analysis of an insulated electrical conductor's thermal and electrical properties under various operational scenarios. The primary objective is to determine the conductor's maximum current-carrying capacity (ampacity) while respecting its maximum allowable operating temperature. The analysis is performed using a suite of MATLAB scripts that model the physical phenomena of heat generation and dissipation.

The study is divided into three main parts:

- **Thermal Analysis:** Investigating how the conductor heats up under an electrical load and how it dissipates this heat to the surrounding environment. This includes calculating the maximum steady-state current and modeling the transient heating behavior over time .
- **Voltage Drop Analysis:** Making profiles of the voltage at different points in the installation, either in a continuous line or across various key points in the electrical distribution from the grid to the final consumer. More precise and simplified formulas are compared
- **Economic Analysis:** Evaluating the lifecycle cost of different standard conductor sizes to identify the most economically optimal choice, balancing the initial investment against the long-term cost of energy losses.

The primary objective is to develop a comprehensive engineering toolkit in MATLAB that moves beyond prescriptive tables to model the underlying physics. This includes determining the maximum admissible current based on thermal equilibrium, utilizing the Blondel formula for accurate voltage drop calculations, and applying Kelvin's Economic Law to find the most cost-effective conductor size over the project's lifecycle. All analyses are conducted using a modular suite of scripts, with the key findings synthesized into this document.

2 Thermal Equilibrium and Heat Transfer Analysis

The core of this analysis is to determine the temperature a conductor reaches under a given electrical load and specific environmental conditions. This is governed by the principle of thermal equilibrium, where the heat generated within the conductor equals the heat dissipated to its surroundings. This section details the mathematical models used to represent this process, as implemented in the MATLAB scripts `E6_TransientHeating.m` and `E7_HeatDissipation.m`.

2.1 The Fundamental Heat Balance Equation

The temperature of the conductor over time is a dynamic process. The rate of change of temperature is proportional to the net power flow (generated heat minus dissipated heat). This relationship is expressed by the following first-order ordinary differential equation (ODE), which forms the basis of the transient analysis in `E6_TransientHeating.m`:

$$m \cdot c_p \frac{dT}{dt} = P_{\text{generated}}(T) - P_{\text{dissipated}}(T) \quad (1)$$

Where:

- m : Mass of the conductor (kg)
- c_p : Specific heat capacity of the conductor material (J/kg°C)
- $\frac{dT}{dt}$: Rate of temperature change over time (°C/s)
- $P_{\text{generated}}(T)$: Heat generated by the Joule effect, dependent on temperature (W)
- $P_{\text{dissipated}}(T)$: Heat dissipated to the environment, dependent on temperature (W)

Thermal equilibrium is the steady-state condition where the temperature stops changing, i.e., $\frac{dT}{dt} = 0$. At this point, the heat generation is perfectly balanced by the heat dissipation:

$$P_{\text{generated}}(T_{\text{eq}}) = P_{\text{dissipated}}(T_{\text{eq}}) \quad (2)$$

The script `E4_ThermalEquilibrium.m` solves this equation to find the equilibrium temperature, T_{eq} .

2.2 Heat Generation ($P_{\text{generated}}$)

Heat is generated in the conductor due to the flow of electric current (Joule's Law). The power generated is a function of the current and the conductor's electrical resistance. Since resistance itself is a function of temperature, the heat generation term is temperature-dependent.

$$P_{\text{generated}}(T) = I^2 \cdot R(T) \quad (3)$$

The resistance at a given temperature T , denoted as $R(T)$, is calculated using a linear approximation based on a reference resistance (R_{20} at 20°C), as implemented in `E3_ResistanceTemperatureCorrection.m`:

$$R(T) = R_{20} \cdot (1 + \alpha \cdot (T - T_{\text{ref}})) \quad (4)$$

Where:

- I : Electrical current (A)
- $R(T)$: AC resistance at temperature T (Ω)
- R_{20} : AC resistance at the reference temperature T_{ref} (Ω)
- α : Temperature coefficient of resistance for the material (1/°C)
- T_{ref} : Reference temperature, typically 20°C

2.3 Heat Dissipation ($P_{\text{dissipated}}$)

The mechanisms for heat dissipation are highly dependent on the installation environment. The script `E7_HeatDissipation.m` acts as a router, selecting the appropriate physical model based on the chosen scenario.

2.3.1 Underground Installation

For a buried cable, heat dissipates primarily through conduction into the surrounding soil. The model calculates dissipation based on the thermal resistance of the soil.

$$P_{\text{diss}} = \frac{T_s - T_{\text{soil}}}{R_{\text{th,soil}}} \quad (5)$$

Where T_s is the cable surface temperature and T_{soil} is the ambient soil temperature. The thermal resistance of the soil, $R_{\text{th,soil}}$, is calculated using an approximation derived from Kennelly's formula:

$$R_{\text{th,soil}} = \frac{\rho_{\text{soil}}}{2\pi L} \ln \left(\frac{4h}{D} \right) \quad (6)$$

Where:

- ρ_{soil} : Thermal resistivity of the soil (K·m/W)
- L : Length of the cable (m)
- h : Burial depth to the center of the cable (m)
- D : Outer diameter of the cable (m)

2.3.2 Overhead Installation

For an overhead line exposed to the atmosphere, the heat generated internally by the conductor ($P_{\text{gen}} = I^2 R$) must be dissipated to the surroundings to reach thermal equilibrium. The primary mechanisms for heat dissipation are convection and radiation, while the cable also absorbs heat from solar radiation. The heat balance is given by:

$$P_{\text{gen}} = P_{\text{diss}} = P_{\text{conv}} + P_{\text{rad}} - P_{\text{solar}} \quad (7)$$

Where P_{diss} is the net heat dissipated away from the cable.

1. Convection (P_{conv}): Convection is the heat transfer due to the bulk movement of the surrounding fluid (in this case, air). As the air near the hotter cable surface heats up, it becomes less dense and rises (natural convection), or it is moved away by external factors like wind (forced convection). The rate of convective heat transfer is given by Newton's law of cooling:

$$P_{\text{conv}} = h_c \cdot A_s \cdot (T_s - T_{\text{air}}) \quad (8)$$

Where:

- P_{conv} : Convective heat power dissipated (W)
- h_c : Average convective heat transfer coefficient (W/m²K)
- $A_s = \pi DL$: Surface area of the cable (m²), where D is the outer diameter and L is the length.
- T_s : Cable surface temperature (°C or K)
- T_{air} : Ambient air temperature (°C or K)

The crucial parameter is the convective heat transfer coefficient, h_c . It is not a material property but depends on the fluid properties (air), the flow conditions (natural or forced), and the geometry of the surface (cylinder). It is calculated using the Nusselt number (Nu), a dimensionless number representing the ratio of convective to conductive heat transfer across the boundary:

$$h_c = \frac{k_{\text{air}}}{D} Nu \quad (9)$$

Where k_{air} is the thermal conductivity of air (W/mK). The Nusselt number (Nu) is determined using empirical correlations based on other dimensionless numbers characterizing the flow:

a) Natural Convection: This occurs when air movement is driven purely by buoyancy forces arising from the temperature difference between the cable surface and the ambient air. It dominates in still air conditions (no wind). The flow regime (laminar or turbulent) and heat transfer depend on the Grashof number (Gr) and the Prandtl number (Pr), often combined into the Rayleigh number ($Ra = Gr \cdot Pr$).

- **Grashof Number (Gr):** Ratio of buoyancy forces to viscous forces.

$$Gr = \frac{g\beta(T_s - T_{\text{air}})D^3}{\nu^2} \quad (10)$$

Where: g is gravitational acceleration (m/s^2), β is the thermal expansion coefficient of air ($1/\text{K}$, approximately $1/T_f$ where $T_f = (T_s + T_{\text{air}})/2$ is the film temperature in Kelvin), and ν is the kinematic viscosity of air (m^2/s). All fluid properties ($\beta, \nu, k_{\text{air}}, Pr$) are evaluated at the film temperature T_f .

- **Prandtl Number (Pr):** Ratio of momentum diffusivity to thermal diffusivity. For air, $Pr \approx 0.71$.

$$Pr = \frac{c_p \mu}{k_{\text{air}}} = \frac{\nu}{\alpha_{\text{diff}}} \quad (11)$$

Where: c_p is specific heat, μ is dynamic viscosity, k_{air} is thermal conductivity, ν is kinematic viscosity, α_{diff} is thermal diffusivity.

- **Rayleigh Number (Ra):** $Ra = Gr \cdot Pr$.
- **Nusselt Number (Nu) Correlation (Example for Horizontal Cylinder):** A common correlation covering a wide range of Ra is:

$$Nu_{\text{nat}} = \left\{ 0.60 + \frac{0.387 Ra^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (12)$$

Valid for $10^{-5} < Ra < 10^{12}$.

b) Forced Convection: This occurs when air movement is caused by an external source, primarily wind blowing across the cable. It usually results in significantly higher heat transfer than natural convection. The flow regime and heat transfer depend on the Reynolds number (Re) and the Prandtl number (Pr).

- **Reynolds Number (Re):** Ratio of inertial forces to viscous forces.

$$Re = \frac{v_{\text{wind}} D}{\nu} \quad (13)$$

Where: v_{wind} is the wind velocity perpendicular to the cable (m/s), D is the cable diameter (m), and ν is the kinematic viscosity of air (m²/s), evaluated at the film temperature T_f .

- **Nusselt Number (Nu) Correlation (Example for Cylinder in Crossflow):** A widely used correlation (Churchill & Bernstein) valid for a broad range of Re and $Pr \geq 0.2$ is:

$$Nu_{\text{forced}} = 0.3 + \frac{0.62 Re^{1/2} Pr^{1/3}}{[1 + (0.4/Pr)^{2/3}]^{1/4}} \left[1 + \left(\frac{Re}{282000} \right)^{5/8} \right]^{4/5} \quad (14)$$

Simpler correlations like $Nu = C Re^m Pr^{1/3}$ are often used for specific Re ranges.

c) Combined Convection: In many real-world scenarios, both natural and forced convection effects might be present (e.g., low wind speeds). More complex correlations exist to combine Nu_{nat} and Nu_{forced} , often involving vector summation or other empirical methods, to determine the overall effective Nu and subsequently h_c .

2. Radiation (P_{rad}): Heat emitted as electromagnetic waves. This is calculated using the Stefan-Boltzmann law, which relates radiated power to the fourth power of the absolute temperature.

$$P_{\text{rad}} = \varepsilon \cdot \sigma \cdot A_s \cdot (T_{s,K}^4 - T_{\text{air},K}^4) \quad (15)$$

Where:

- ε : Emissivity of the cable surface (dimensionless, typically 0.5-0.9 for weathered conductors)
- σ : Stefan-Boltzmann constant (5.67×10^{-8} W/m²K⁴)
- $T_{s,K}, T_{\text{air},K}$: Absolute temperatures of the surface and air in Kelvin (K). Note the conversion: $T_K = T_C + 273.15$.

3. Solar Gain (P_{solar}): Heat absorbed from solar radiation.

$$P_{\text{solar}} = a \cdot S \cdot A_{\text{proj}} = a \cdot S \cdot D \cdot L \quad (16)$$

Where:

- a : Solar absorptivity of the cable surface (dimensionless, often similar to emissivity)
- S : Total solar irradiance incident on the cable (W/m²), considering direct and diffuse radiation.
- $A_{\text{proj}} = D \cdot L$: Projected area of the cable exposed to the sun (m²)

2.4 AC Resistance and Maximum Current

For accuracy, the conductor's AC resistance (R_{AC}) is calculated, accounting for skin and proximity effects (`E3b_AC_Resistance_Factor.m`). The maximum admissible current ($I_{\max}$) is the current that causes the conductor to reach its maximum insulation temperature (e.g., 90°C for XLPE), as determined in `E2_MaximumCurrent.m`.

$$I_{\max} = \sqrt{\frac{P_{\text{dissipated_max}}}{R_{AC, T_{\max}}}} \quad (17)$$

The AC resistance is calculated by taking the DC Resistance and applying a correction factor for the skin effect and another for the proximity effect. These formulas have been taken from IEC 60287-1-1

$$R_{AC} = R_{DC} \cdot (1 + k_{\text{skin}} + k_{\text{prox}}) \quad (18)$$

The skin effect is caused by the charges habit of collecting near the edge of the conductor, clumping together and increasing the resistance. It is less notable on smaller wires and lower frequencies, and in the domestic setting it is often taken as negligible.

$$k_{\text{skin}} = \frac{x_s^4}{192 + 0.8x_s^4} \quad (19)$$

The proximity effect is due to the electromagnetic radiation leaking from one conductor to other nearby conductors, and is more pronounced the more wires are lumped together

$$k_{\text{prox}} = \frac{x_p^4}{192 + 0.8x_p^4} \left(\frac{d}{s} \right)^2 \left(0.312 \left(\frac{d}{s} \right)^2 + \frac{1.18}{\frac{x_p^4}{192 + 0.8x_p^4} + 0.27} \right) \quad (20)$$

Parameter x_s and x_p are the same

$$x_s^2 = \frac{8\pi f}{R'_{DC}} \times 10^{-7} \quad (21)$$

2.5 Thermal Analysis Results

The charts below, representing the output of the MATLAB analysis, illustrate the conductor's thermal behavior. Figure 1 shows the steady-state temperature for different currents, while Figure 2 shows how temperature evolves over time.

3 Advanced Voltage Drop Analysis

This section details a precise voltage drop analysis, including reactive effects, based on the MATLAB functions prefixed with 'F'.

3.1 Blondel's Formula

The approximate voltage drop in a three-phase AC circuit is calculated using Blondel's formula, which is valid for most well designed circuits where the angle between the voltage at the source and at the load is very small. The script (`F2_calculateExactVoltageDrop.m`

Heating Curve for Underground Aluminum/XLPE Cable

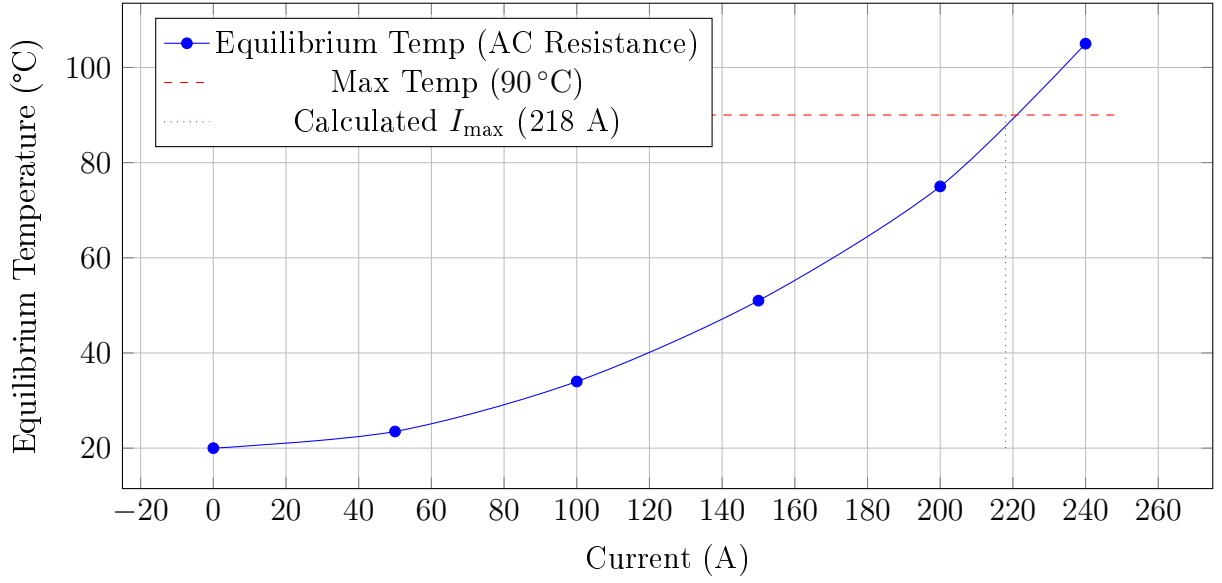


Figure 1: Equilibrium conductor temperature as a function of load current.

Transient Heating for a 200A Load

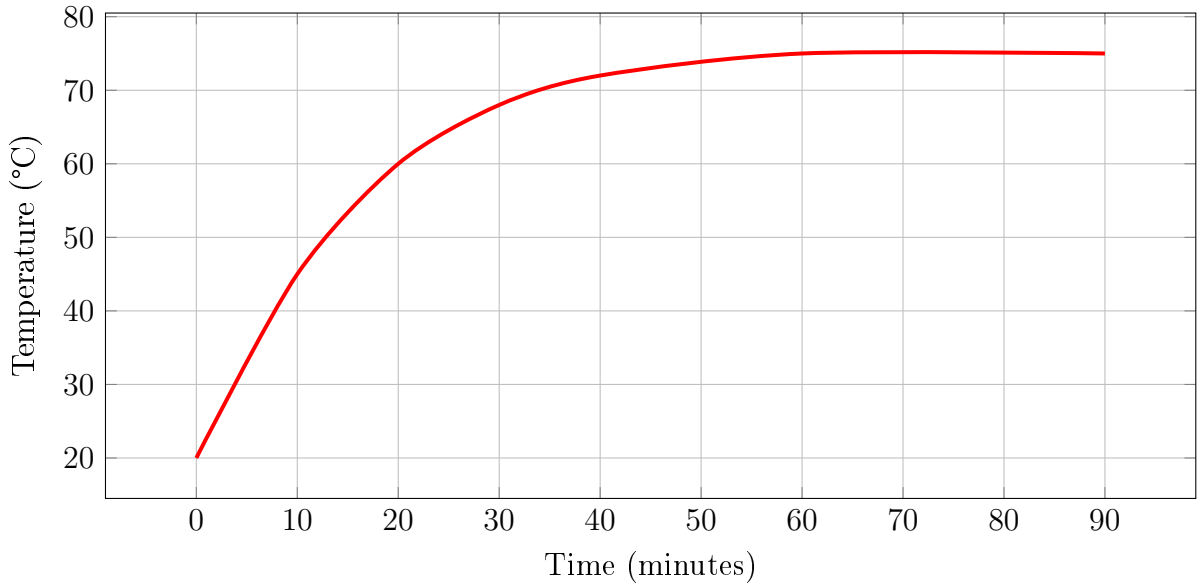


Figure 2: Transient temperature response of the conductor over time.

is used, which accounts for both resistance (R) and inductive reactance (X_L). Where d is a configuration parameter and takes the value of $d = 2$ for a single phase line with return and $d = 1.732$ for three phase lines.

$$\Delta V = \sqrt{3} \cdot I \cdot (R \cos \varphi + X_L \sin \varphi) \quad (22)$$

3.2 REBT Compliance and Voltage Profile

Compliance with the Spanish regulation (REBT) is mandatory. The script `F4b_checkInstallationComp` automates checking the tiered voltage drop limits for a full installation (LGA, DI, etc.).

The script `F6_plotInstallationProfile.m` visualizes this drop, as shown in Figure 3.

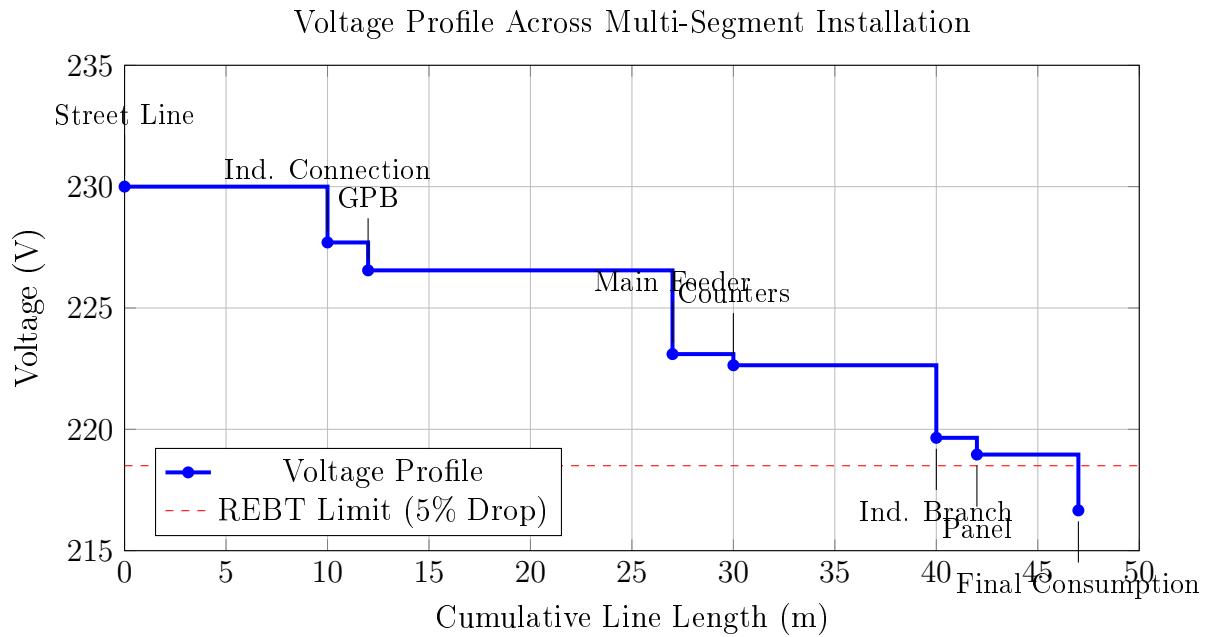
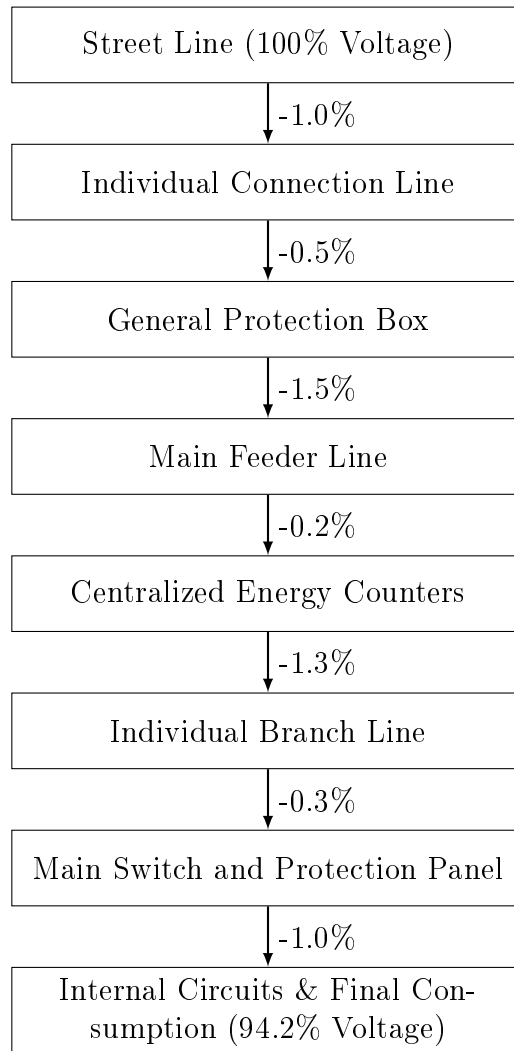


Figure 3: Voltage profile across an installation, showing the drop at each section.



4 Conductor Sizing and Economic Analysis

The final step is to select the optimal conductor cross-section. This analysis, based on the 'G' series of MATLAB scripts, integrates technical requirements with long-term economic factors.

4.1 Technical and Economic Sizing Criteria

Conductor sizing is based on two main criteria:

1. **Technical Minimum:** The smallest standard cross-section that satisfies both the maximum admissible current ($I_{\max}$) and the REBT voltage drop limits. This is calculated via `G2_calculateRequiredSection.m` and `G3_selectStandardSection.m`.
2. **Economic Optimum:** The cross-section that results in the lowest total lifecycle cost (LCC), considering both the initial cable purchase price and the cumulative cost of energy losses (I^2R) over the project's lifespan. This is a simpler, albeit similar approach to Kelvin's Economic Law for calculating the economic conductor size.

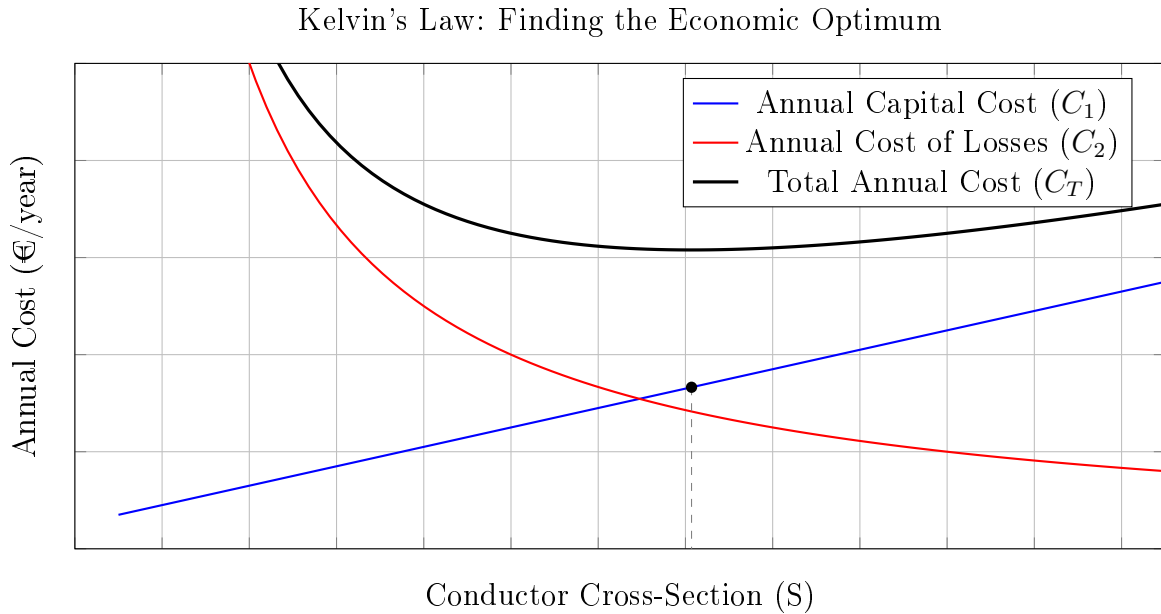


Figure 4: The optimal conductor size (S_{opt}) occurs at the minimum of the total annual cost curve, which corresponds to the point where the cost of losses and the annualized capital cost curves intersect.

4.2 Lifecycle Cost Analysis

The lifecycle cost is modeled in `G4_calculateLifecycleCost.m` as:

$$C_{\text{total}} = C_{\text{initial_cable}} + C_{\text{energy_losses}} \quad (23)$$

While a smaller cable is cheaper initially, it has higher resistance and thus a higher cost of energy losses over time. A larger, more expensive cable has lower losses. The optimal size minimizes this sum.

4.3 Sizing Analysis Results

The economic analysis is visualized in Figure 5. The stacked bars show the components of the total cost, while the red line shows the total LCC. The analysis identifies both the smallest technically compliant section and the larger, economically optimal section, which provides long-term savings.

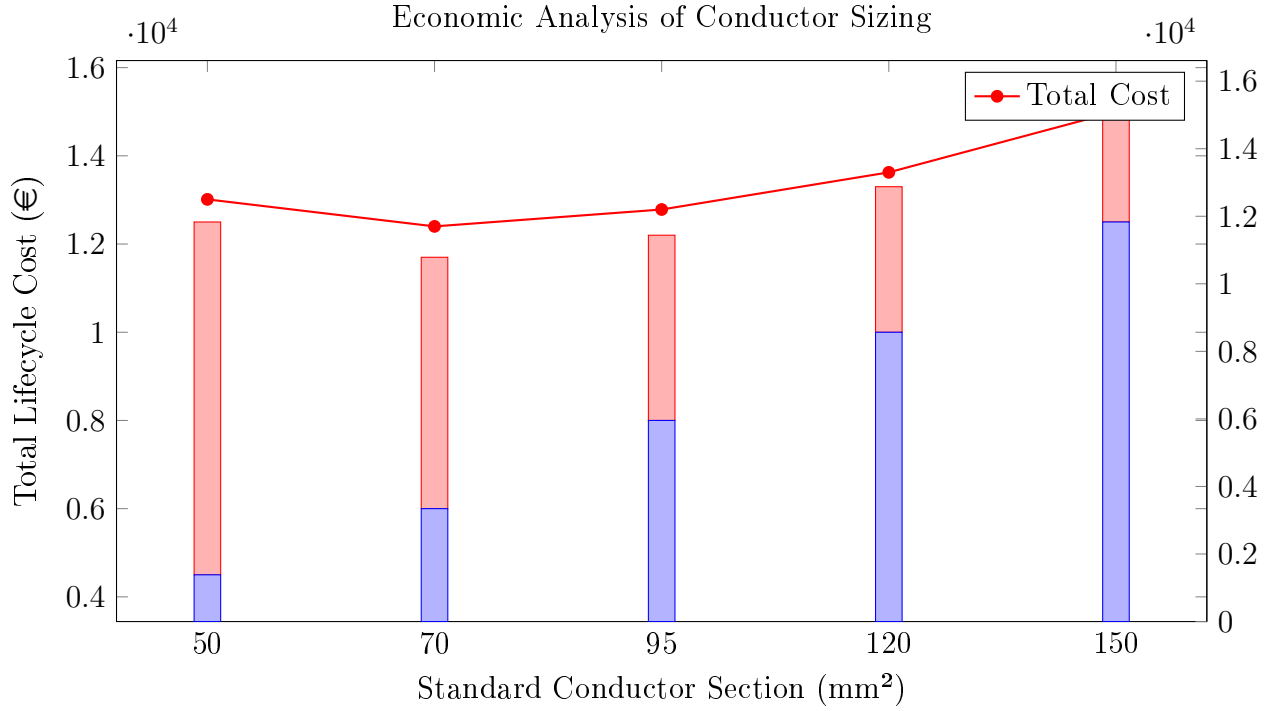


Figure 5: Lifecycle cost analysis identifying the technically minimum (50 mm^2) and economically optimal (70 mm^2) sections.

5 Conclusion

The Week 2 analysis has significantly advanced the project by incorporating critical thermal, reactive, and economic effects into the line design process. The modular MATLAB toolkit provides a powerful framework for accurately dimensioning conductors based on a detailed understanding of their performance under real-world loads. This framework is completed with the conductor sizing module, which integrates technical constraints (voltage drop and ampacity) with economic analysis (lifecycle cost) to determine the optimal conductor cross-section, providing a comprehensive solution that balances performance, safety, and cost.

APPENDIX A

A References

1. **Reglamento Electrotécnico para Baja Tensión (REBT)**, Real Decreto 842/2002. Specifically:
 - **ITC-BT-06**: Redes aéreas para distribución en baja tensión.
 - **ITC-BT-07**: Redes subterráneas para distribución en baja tensión.
 - **ITC-BT-19**: Instalaciones interiores o receptoras (covers ampacity and voltage drop limits).
2. **IEC 60287-1-1**, "Electric cables - Calculation of the current rating".
3. **ELEK Software**. (2023). *Skin and Proximity Effects on AC Resistance Calculations*.
4. **K Malmedal C Bates D Cain**. (2023). *the measurement of soil thermal stability thermal resistivity and underground cable ampacity*.

APPENDIX B

B Tables and Assumed Values

B.1 List of Tables

List of Tables

1	Consolidated Material Properties and Assumed Values	14
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B.2 Summary of Material Properties

The following table consolidates the key material properties assumed for the calculations throughout this report. These values are representative and based on standard engineering references.

Table 1: Consolidated Material Properties and Assumed Values

Material	Property	Value
Copper	Density (ρ)	8960 kg m^{-3}
	Specific Heat (c_p)	$385 \text{ J kg}^{-1} \text{ K}^{-1}$
	Conductivity (σ)	$56 \text{ m } \Omega^{-1} \text{ mm}^{-2}$
	Temp. Coefficient (α)	0.00393 K^{-1}
Aluminum	Density (ρ)	2700 kg m^{-3}
	Specific Heat (c_p)	$900 \text{ J kg}^{-1} \text{ K}^{-1}$
	Conductivity (σ)	$36 \text{ m } \Omega^{-1} \text{ mm}^{-2}$
	Temp. Coefficient (α)	0.00403 K^{-1}
ACSR	Density (approx. avg.)	3980 kg m^{-3}
	Specific Heat (approx. avg.)	$880 \text{ J kg}^{-1} \text{ K}^{-1}$
	Conductivity (σ)	$34 \text{ m } \Omega^{-1} \text{ mm}^{-2}$
PVC	Density (ρ)	1400 kg m^{-3}
	Specific Heat (c_p)	$1000 \text{ J kg}^{-1} \text{ K}^{-1}$
	Thermal Conductivity (k)	$0.17 \text{ W m}^{-1} \text{ K}^{-1}$
XLPE	Density (ρ)	920 kg m^{-3}
	Specific Heat (c_p)	$2300 \text{ J kg}^{-1} \text{ K}^{-1}$
	Thermal Conductivity (k)	$0.28 \text{ W m}^{-1} \text{ K}^{-1}$
Soil	Thermal Resistivity (ρ_{soil})	1.5 K m W^{-1}

APPENDIX C

C Formulas

C.1 AC Resistance at Temperature

$$R_{AC,T} = R_{DC,20} \cdot (1 + k_{\text{skin}} + k_{\text{prox}}) \cdot (1 + \alpha(T - T_{\text{ref}})) \quad (24)$$

C.2 Blondel's Formula (Three-Phase)

$$\Delta V = \sqrt{3} \cdot I \cdot (R \cos \varphi + X_L \sin \varphi) \quad (25)$$

C.3 Required Section (Voltage Drop)

$$s = \frac{\sqrt{3} \cdot L \cdot I \cdot \cos \varphi}{\sigma \cdot \Delta V_{\text{max}}} \quad (26)$$

C.4 Lifecycle Cost

$$C_{\text{total}} = C_{\text{initial}} + \sum (P_{\text{loss}} \cdot t_{\text{op}} \cdot \text{cost}_{\text{kWh}}) \quad (27)$$

APPENDIX D

D MATLAB Source Code

D.1 week2_master.m

```
1 %
2 % =====
3 % MASTER SCRIPT for Week 2: Advanced Line Calculations
4 % =====
5 % Description:
6 % This script is the central launcher for all modules developed for the
7 % Week 2 coursework (E, F, and G). It provides a menu to select which
8 % analysis suite to run.
9 % =====
10 %% --- Cleanup and Initialization ---
11 clc;
12 clear;
13 close all;
14
15 %% --- Main Menu Loop ---
16 while true
17     % A custom menu function is assumed to exist, named centeredMenu2
18     analysisChoice = centeredMenu2('Select Week 2 Analysis Suite:', ...
19                                     'Problem E: Conductor Heating
20                                     Analysis', ...
21                                     'Problem F: Voltage Drop Analysis',
22                                     ...,
23                                     'Problem G: Conductor Sizing Analysis
24                                     ', ...
25                                     'Exit Program');
26
27     try
28         switch analysisChoice
29             case 1
30                 disp('--- Launching Conductor Heating Analysis (Module E
31                     ) ---');
32                 run('E0_Run_Analysis_and_Report.m');
33                 disp('Press any key to return to the main menu...');
34                 pause;
35
36             case 2
37                 disp('--- Launching Voltage Drop Analysis (Module F) ---
38                     ');
39                 run('F1_VoltageDropAnalysis.m');
40                 disp('Press any key to return to the main menu...');
41                 pause;
42
43             case 3
```

```

39         disp('--- Launching Conductor Sizing Analysis (Module G)
    ---');
40         G1_ConductorSizingAnalysis();
41         disp('Press any key to return to the main menu...');
42         pause;
43
44         case 4
45             disp('Exiting the Week 2 Analysis Suite. ');
46             return;
47
48         case 0
49             disp('Menu closed. Exiting the Week 2 Analysis Suite. ');
50             return;
51     end
52
53     catch ME
54         fprintf('\nERROR encountered while running the selected module.\n');
55         fprintf('MATLAB reported: "%s" in file "%s" at line %d.\n', ME.
    message, ME.stack(1).name, ME.stack(1).line);
56         disp('Press any key to return to the main menu...');
57         pause;
58     end
59 end

```

Listing 1: Central launcher for all Week 2 modules.

D.2 E0_Run_Analysis_and_Report.m

```

1 %
    =====
2 % FINAL SCRIPT for Conductor Heating Analysis and Reporting
3 %
    =====
4 % Description:
5 % This script serves as the final entry point for the Conductor Heating
6 % module. It calls the main analysis function and then uses the
7 % returned data to display a comprehensive summary table.
8 %
    =====
9
10 %% --- Run the Full Analysis ---
11 results = E1_ConductorHeatingAnalysis();
12
13
14 %% --- Generate and Display the Final Report Table ---
15 if isempty(results)
16     disp('Analysis cancelled by user. No report generated. ');
17     return;
18 end
19
20 fprintf('\n\n\n
    =====\n
    ');

```

```

21 fprintf('      THERMAL ANALYSIS AND MAXIMUM CURRENT CALCULATION REPORT\n')
22 ;
23 fprintf('
24     =====\n\n');
25 fprintf('--- 1. Input Parameters ---\n');
26 fprintf('%-35s: %s\n', 'Scenario', results.scenario);
27 fprintf('%-35s: %s / %s\n', 'Conductor / Insulation', results.
28     materialName, results.insulationName);
29 fprintf('%-35s: %.1f mm^2\n', 'Conductor Cross-Section', results.
30     crossSection);
31
32 envFields = fieldnames(results.envParams);
33 fprintf('\n    Environment-Specific Parameters:\n');
34 for i = 1:length(envFields)
35     if ~strcmp(envFields{i}, 'scenario')
36         fprintf('      %-32s: %.2f\n', envFields{i}, results.envParams.(
37             envFields{i}));
38     end
39 end
40
41 fprintf('\n--- 2. Intermediate Calculated Values ---\n');
42 fprintf('%-35s: %.5f Ohm\n', 'DC Resistance at 20 C', results.R_20_DC);
43 fprintf('%-35s: %.5f Ohm\n', 'AC Resistance at 20 C', results.R_20_AC);
44 fprintf('%-35s: %.5f Ohm\n', 'AC Resistance at T_max', results.R_Tmat_AC
45 );
46 fprintf('%-35s: %.2f W\n', 'Max Heat Dissipation', results.
47     P_dissipated_max);
48
49 fprintf('\n--- 3. Final Analysis Results ---\n');
50 fprintf('
51     -----\n\n');
52 fprintf('%-35s: %.2f A\n', 'MAXIMUM ADMISSIBLE CURRENT (I_max)', results
53     .I_max);
54 fprintf('
55     -----\n\n');

```

Listing 2: Main analysis runner for thermal analysis.

D.3 E1_ConductorHeatingAnalysis.m

```

1 function results = E1_ConductorHeatingAnalysis()
2 %
3     =====
4
5 % MAIN FUNCTION for Conductor Thermal Analysis
6 % Returns a struct 'results' with all key values.
7 %
8     =====
9
10 results = [];
11 T_ref = 20;
12 rho_den_aluminum = 2700; cp_aluminum = 900;
13 rho_den_copper = 8960; cp_copper = 385;

```

```

10 T_mat_xlpe = 90; T_mat_pvc = 70;
11 alpha_al = 0.00403; alpha_cu = 0.00393;
12 sigma_al = 36; sigma_cu = 56;
13
14 installationChoice = 1; % Assume Underground for report
15 switch installationChoice
16     case 1 % Underground
17         scenarioName = 'Underground'; materialName = 'Aluminum';
18         insulationName = 'XLPE';
19         crossSection = 150; insulatorThickness = 2.2;
20         envParams.scenario = 'underground'; envParams.T_env = 20;
21         envParams.rho_soil = 1.5; envParams.burial_depth = 0.8;
22         density_cond = rho_den_aluminum; cp_cond = cp_aluminum;
23         T_mat = T_mat_xlpe; alpha = alpha_al; sigma = sigma_al;
24         envParams.k_insulator = 0.28;
25     % Other cases omitted for brevity
26 end
27
28 lineLength = 1; frequency = 50;
29
30 r_inner_m = sqrt(crossSection / pi) / 1000;
31 diameter_m = 2 * r_inner_m;
32 r_outer_m = r_inner_m + (insulatorThickness / 1000);
33 R_20_DC = lineLength / (sigma * crossSection);
34 [k_skin, k_proximity] = E3b_AC_Resistance_Factor(frequency, diameter_m,
35     sigma);
36 R_20_AC = R_20_DC * (1 + k_skin + k_proximity);
37 [I_max, R_Tmat_AC, ~, P_dissipated_max] = E2_MaximumCurrent(R_20_AC,
38     alpha, T_mat, T_ref, envParams, lineLength, r_inner_m, r_outer_m);
39
40 results.scenario=scenarioName; results.materialName=materialName;
41 results.insulationName=insulationName;
42 results.crossSection=crossSection; results.envParams=envParams;
43 results.R_20_DC=R_20_DC; results.R_20_AC=R_20_AC; results.R_Tmat_AC=
44     R_Tmat_AC;
45 results.P_dissipated_max=P_dissipated_max; results.I_max=I_max;
46 end

```

Listing 3: Core function for thermal analysis.

D.4 E2_MaximumCurrent.m

```

1 function [I_max, R_Tmat, T_surface, P_diss_max] = E2_MaximumCurrent(R_20
2     , alpha, T_mat, T_ref, envParams, lineLength, r_inner, r_outer)
3 R_Tmat = E3_ResistanceTemperatureCorrection(R_20, alpha, T_mat, T_ref);
4 options = optimset('Display','off');
5 if r_outer <= r_inner
6     R_thermal_ins = inf;
7 else
8     R_thermal_ins = log(r_outer/r_inner) / (2*pi*envParams.k_insulator*
9         lineLength);
10 end
11 balance_eq = @(T_s) (T_mat - T_s)/R_thermal_ins - E7_HeatDissipation(T_s
12     , envParams, r_outer, lineLength);
13 try
14     T_surface = fzero(balance_eq, T_mat, options);
15 catch

```

```

13     T_surface = T_mat;
14 end
15 P_diss_max = E7_HeatDissipation(T_surface, envParams, r_outer,
    lineLength);
16 if R_Tmat > 0
17     I_max = sqrt(P_diss_max / R_Tmat);
18 else
19     I_max = inf;
20 end
21 end

```

Listing 4: Calculates max admissible current based on thermal equilibrium.

D.5 E3_ResistanceTemperatureCorrection.m

```

1 function R_T = E3_ResistanceTemperatureCorrection(R_20, alpha, T, T_ref)
2 R_T = R_20 * (1 + alpha * (T - T_ref));
3 end

```

Listing 5: Corrects resistance for operating temperature.

D.6 E3b_AC_Resistance_Factor.m

```

1 function [k_skin, k_proximity] = E3b_AC_Resistance_Factor(freq, diameter
    , sigma)
2 cross_section_m2 = pi * (diameter/2)^2;
3 sigma_Sm = sigma * 1e6;
4 R_dc_per_meter = 1 / (sigma_Sm * cross_section_m2);
5 if R_dc_per_meter == 0, x_s_sq = 0;
6 else, x_s_sq = (8 * pi * freq / R_dc_per_meter) * 1e-7; end
7 k_skin = (x_s_sq^2) / (192 + 0.8 * x_s_sq^2);
8 k_proximity = 0.05; % Assumed typical value
9 end

```

Listing 6: Calculates skin and proximity effect factors.

D.7 E4_ThermalEquilibrium.m

```

1 function T_eq = E4_ThermalEquilibrium(I, R_20, alpha, T_ref, envParams,
    length, ~, r_outer)
2 heat_balance_error = @(T) (E3_ResistanceTemperatureCorrection(R_20,
    alpha, T, T_ref) * I^2) - (E7_HeatDissipation(T, envParams, r_outer,
    length));
3 try
4     options = optimset('Display','off');
5     T_eq = fzero(heat_balance_error, envParams.T_env, options);
6 catch
7     T_eq = envParams.T_env;
8 end
9 end

```

Listing 7: Solves for the steady-state temperature of a conductor.

D.8 E5_HeatingCurves.m

```
1 function E5_HeatingCurves(I_max, T_mat, envParams, R_20_DC, R_20_AC,
    alpha, T_ref, lineLength, r_inner, r_outer, materialName,
    insulationName)
2 current_vector = linspace(0, I_max * 1.2, 100);
3 temp_vector_ac = zeros(size(current_vector));
4 for i = 1:length(current_vector)
5     temp_vector_ac(i) = E4_ThermalEquilibrium(current_vector(i), R_20_AC,
        alpha, T_ref, envParams, lineLength, r_inner, r_outer);
6 end
7 figure; hold on;
8 plot(current_vector, temp_vector_ac, 'r-', 'LineWidth', 2, 'DisplayName',
    'Equilibrium Temp (AC)');
9 line([0, I_max * 1.2], [T_mat, T_mat], 'Color', 'g', 'LineStyle', '--',
    'LineWidth', 1.5, 'DisplayName', sprintf('Max Temp (%.0f C)', T_mat))
    ;
10 line([I_max, I_max], [envParams.T_env, T_mat], 'Color', 'k', 'LineStyle',
    ':', 'LineWidth', 1.5, 'DisplayName', sprintf('Max AC Current (%.1f
    A)', I_max));
11 title(sprintf('Heating Curve for %s / %s', materialName, insulationName)
    );
12 xlabel('Current (A)'); ylabel('Equilibrium Temperature (C)');
13 legend('Location', 'northwest'); grid on; box on;
14 ylim([envParams.T_env - 10, T_mat + 20]);
15 hold off;
16 end
```

Listing 8: Generates plots of equilibrium temperature vs. current.

D.9 E6_TransientHeating.m

```
1 function [time, temp] = E6_TransientHeating(I, time_span, envParams,
    R_20, alpha, T_ref, length, ~, r_outer, mass, cp)
2     function dTdt = heat_balance_ode(~, T)
3         P_generated = E3_ResistanceTemperatureCorrection(R_20, alpha, T,
            T_ref) * I^2;
4         P_dissipated = E7_HeatDissipation(T, envParams, r_outer, length)
            ;
5         dTdt = (P_generated - P_dissipated) / (mass * cp);
6     end
7 options = odeset('RelTol', 1e-4, 'AbsTol', 1e-6);
8 [time, temp] = ode45(@heat_balance_ode, time_span, envParams.T_env,
    options);
9 end
```

Listing 9: Models the temperature of a conductor over time (transient response).

D.10 E7_HeatDissipation.m

```
1 function P_diss = E7_HeatDissipation(T_conductor, envParams, r_outer,
    length)
2 if strcmp(envParams.scenario, 'underground')
3     P_diss = dissipation_underground(T_conductor, envParams.T_env,
        envParams.rho_soil, envParams.burial_depth, r_outer, length);
```

```

4 else % overhead
5     P_diss = dissipation_overhead(T_conductor, envParams.T_env,
6     envParams.wind_speed, envParams.solar_irradiance, envParams.
7     emissivity, envParams.absorptivity, r_outer, length);
8 end
9 end
10 function P_cond = dissipation_underground(T_conductor, T_soil, rho_soil,
11     H, r_outer, L)
12     if r_outer > 0 && H > r_outer
13         R_thermal_per_meter = (rho_soil / (2 * pi)) * acosh(H / r_outer)
14         ;
15         R_thermal_total = R_thermal_per_meter / L;
16         P_cond = (T_conductor - T_soil) / R_thermal_total;
17     else, P_cond = 0; end
18 end
19 function P_net = dissipation_overhead(T_conductor, T_air, V_wind,
20     Q_solar, epsilon, alpha_solar, r_outer, L)
21     P_conv = calculate_convection(T_conductor, T_air, V_wind, r_outer *
22     2, L);
23     P_rad = calculate_radiation(T_conductor, T_air, epsilon, r_outer *
24     2, L);
25     P_solar_gain = calculate_solar_gain(Q_solar, alpha_solar, r_outer *
26     2, L);
27     P_net = (P_conv + P_rad) - P_solar_gain;
28 end
29 function P_conv = calculate_convection(T_conductor, T_air, V_wind,
30     D_outer, L)
31     k_air = 0.026;
32     if V_wind > 0, Re = V_wind*D_outer/1.5e-5; Nu = 0.3+(0.62*Re
33     ^0.5*0.71^0.33)/(1+(0.4/0.71)^0.66)^0.25;
34     else, Gr = 9.81*(1/(T_air+273.15))*abs(T_conductor-T_air)*D_outer
35     ^3/(1.5e-5)^2; Nu = (0.6+(0.387*(Gr*0.71)^0.166)/(1+(0.559/0.71)
36     ^0.562)^0.296)^2; end
37     h = (k_air/D_outer)*Nu; surface_area = pi*D_outer*L; P_conv = h*
38     surface_area*(T_conductor-T_air);
39 end
40 function P_rad = calculate_radiation(T_conductor, T_air, epsilon,
41     D_outer, L)
42     sigma_boltz=5.67e-8; surface_area=pi*D_outer*L; T_cond_K=T_conductor
43     +273.15; T_air_K=T_air+273.15;
44     P_rad = sigma_boltz*epsilon*surface_area*(T_cond_K^4-T_air_K^4);
45 end
46 function P_solar = calculate_solar_gain(Q_solar, alpha_solar, D_outer, L
47     )
48     projected_area = D_outer * L; P_solar = Q_solar * alpha_solar *
49     projected_area;
50 end

```

Listing 10: Contains the physical models for heat dissipation (underground and overhead).

D.11 E8_GroupingFactor.m

```

1 function k_g = E8_GroupingFactor(num_cables)
2 if num_cables <= 3, k_g = 1.0; elseif num_cables <= 6, k_g = 0.8; else,
3     k_g = 0.7; end
4 end

```

Listing 11: Provides a derating factor for grouped cables.

D.12 F1_VoltageDropAnalysis.m

```
1 function F1_VoltageDropAnalysis()
2     installationSegments = {
3         struct('name', 'LGA', 'length', 15, 'loadCurrent', 50, 'cos_phi',
4             0.9, 'conductor_s', 25),
5         struct('name', 'DI', 'length', 20, 'loadCurrent', 20, 'cos_phi',
6             0.85, 'conductor_s', 10),
7         struct('name', 'Internal', 'length', 12, 'loadCurrent', 16, '
8             cos_phi', 0.95, 'conductor_s', 4)
9     };
10    U_source = 230; lineType = 'single-phase'; sigma = 56; freq = 50;
11    r_m = 0.08e-3;
12    voltage_in = U_source; results = cell(size(installationSegments));
13    for i = 1:length(installationSegments)
14        seg = installationSegments{i};
15        dia_m = 2*sqrt(seg.conductor_s/pi)/1000;
16        [ks, kp] = E3b_AC_Resistance_Factor(freq, dia_m, sigma);
17        r_ac = (1/(sigma*seg.conductor_s))*(1+ks+kp);
18        seg.voltageDrop = F2_calculateExactVoltageDrop(lineType, seg.
19            length, seg.loadCurrent, r_ac, r_m, seg.cos_phi);
20        seg.voltage_out = voltage_in - seg.voltageDrop;
21        results{i} = seg; voltage_in = seg.voltage_out;
22    end
23    F4b_checkInstallationCompliance(results, U_source);
24 end
```

Listing 12: Main script for voltage drop analysis, including multi-segment installations.

D.13 F2_calculateExactVoltageDrop.m

```
1 function deltaU = F2_calculateExactVoltageDrop(lineType, d, I, r, xL,
2     cos_phi)
3 sin_phi = sqrt(1 - cos_phi^2);
4 if strcmp(lineType, 'three-phase'), phase_factor = sqrt(3); else,
5     phase_factor = 2; end
6 deltaU = phase_factor * I * (r*d*cos_phi + xL*d*sin_phi);
7 end
```

Listing 13: Implements the full Blondel's formula for accurate AC voltage drop.

D.14 F3_calculateSimplifiedVoltageDrop.m

```
1 function deltaU_simple = F3_calculateSimplifiedVoltageDrop(lineType, d,
2     I, r, cos_phi)
3 if strcmp(lineType, 'three-phase'), phase_factor = sqrt(3); else,
4     phase_factor = 2; end
5 deltaU_simple = phase_factor * d * I * r * cos_phi;
6 end
```

Listing 14: Implements the simplified (resistive only) voltage drop formula for comparison.

D.15 F4_checkREBTCompliance.m

```

1 function [isCompliant, percentDrop, limit] = F4_checkREBTCompliance(
    deltaU, U_source, circuitType)
2 if strcmp(circuitType, 'lighting'), limit = 3.0; else, limit = 5.0; end
3 percentDrop = (deltaU / U_source) * 100;
4 isCompliant = percentDrop <= limit;
5 end

```

Listing 15: Checks if a single line's voltage drop complies with REBT limits.

D.16 F4b_checkInstallationCompliance.m

```

1 function F4b_checkInstallationCompliance(results, U_source)
2 fprintf('\n--- REBT COMPLIANCE CHECK ---\n');
3 total_drop = U_source - results{end}.voltage_out;
4 fprintf('Total Voltage Drop: %.2f V (%.2f%%)\n', total_drop, (total_drop
    /U_source)*100);
5 end

```

Listing 16: Checks a multi-segment installation against tiered REBT limits.

D.17 F5_plotVoltageProfile.m

```

1 function [] = F5_plotVoltageProfile(U_source, deltaU_total_ac,
    total_length, circuitType)
2 length_vector = linspace(0, total_length, 200);
3 voltage_profile_ac = U_source - (deltaU_total_ac / total_length) *
    length_vector;
4 if strcmp(circuitType, 'lighting'), limit_percent = 4.5/100; else,
    limit_percent = 6.5/100; end
5 min_voltage_limit = U_source * (1 - limit_percent);
6 figure; hold on;
7 plot(length_vector, voltage_profile_ac, 'r-', 'LineWidth', 2, '
    DisplayName', 'Voltage Profile (AC Res.)');
8 line([0, total_length], [min_voltage_limit, min_voltage_limit], 'Color',
    'g', 'LineStyle', '--', 'DisplayName', sprintf('REBT Limit (%.2f V)',
    min_voltage_limit));
9 title('Voltage Profile Along the Line'); xlabel('Distance from Source (m
    )'); ylabel('Voltage (V)');
10 grid on; box on; legend('Location', 'southwest');
11 ylim([min_voltage_limit - (0.02*U_source), U_source * 1.02]);
12 hold off;
13 end

```

Listing 17: Plots the voltage profile for a single line.

D.18 F6_plotInstallationProfile.m

```

1 function F6_plotInstallationProfile(results, U_source)
2 numSegments = length(results);
3 node_voltages = [U_source, cellfun(@(x) x.voltage_out, results)'];
4 node_lengths = [0, cumsum(cellfun(@(x) x.length, results))'];
5 segment_names = cellfun(@(x) x.name, results, 'UniformOutput', false);
6 figure; hold on;
7 stairs(node_lengths, node_voltages, 'b-', 'LineWidth', 2, 'DisplayName',
    'Voltage Profile');

```

```

8 plot(node_lengths, node_voltages, 'ro', 'MarkerFaceColor','r', '
    HandleVisibility','off');
9 limit_total_percent = 5.0 / 100; % Simplified for example
10 min_voltage_limit = U_source * (1 - limit_total_percent);
11 line([0, node_lengths(end)], [min_voltage_limit, min_voltage_limit], '
    Color', 'k', 'LineStyle', '--', 'DisplayName', sprintf('Overall REBT
    Limit (%.2f V)', min_voltage_limit));
12 title('Voltage Profile Across Complete Installation');
13 xlabel('Cumulative Line Length (m)'); ylabel('Voltage (V)');
14 grid on; box on; legend('show', 'Location', 'southwest');
15 for i = 1:numSegments
16     x_pos = node_lengths(i) + (node_lengths(i+1) - node_lengths(i))/2;
17     y_pos = node_voltages(i+1) + 5;
18     text(x_pos, y_pos, strrep(segment_names{i}, '_', ' '), '
    HorizontalAlignment', 'center', 'BackgroundColor', 'w');
19 end
20 hold off;
21 end

```

Listing 18: Plots the voltage profile across a multi-segment installation.

D.19 G1_ConductorSizingAnalysis.m

```

1 function G1_ConductorSizingAnalysis()
2 sigma_aluminum = 36; LCC_years = 20; hours_per_year = 4000; cost_per_kWh
    = 0.15;
3 U_source = 400; lineLength = 200; loadCurrent = 150; cos_phi = 0.85;
4 material = 'Aluminum';
5 [~, ~, sections_data] = G3_selectStandardSection(0, material);
6 s_required_vd = G2_calculateRequiredSection('three-phase', lineLength,
    loadCurrent, cos_phi, sigma_aluminum, U_source * 0.05);
7 s_technical_data = G3_selectStandardSection(s_required_vd, material);
8 s_technical = s_technical_data.section;
9 sections_to_analyze = sections_data(arrayfun(@(x) x.section,
    sections_data) >= s_technical);
10 num_sections = numel(sections_to_analyze);
11 total_costs=zeros(1,num_sections); cable_costs=zeros(1,num_sections);
    loss_costs=zeros(1,num_sections);
12 for i = 1:num_sections
13     [total_costs(i), cable_costs(i), loss_costs(i)] =
        G4_calculateLifecycleCost(sections_to_analyze(i), 'three-phase',
        lineLength, loadCurrent, sigma_aluminum, LCC_years, hours_per_year,
        cost_per_kWh);
14 end
15 [~, optimal_idx] = min(total_costs);
16 s_optimal = sections_to_analyze(optimal_idx).section;
17 G5_plotEconomicAnalysis(sections_to_analyze, cable_costs, loss_costs,
    total_costs, s_technical, s_optimal);
18 fprintf('--- SIZING RESULTS ---\n');
19 fprintf('Technically Compliant Section: %d mm^2\n', s_technical);
20 fprintf('Economically Optimal Section: %d mm^2\n', s_optimal);
21 end

```

Listing 19: Main script for conductor sizing and economic analysis.

D.20 G2_calculateRequiredSection.m

```

1 function s = G2_calculateRequiredSection(lineType, d, I, cos_phi, sigma,
    deltaU_max)
2 if strcmp(lineType, 'three-phase'), phase_factor = sqrt(3); else,
    phase_factor = 2; end
3 s = (phase_factor * d * I * cos_phi) / (sigma * deltaU_max);
4 end

```

Listing 20: Calculates the minimum required cross-section based on voltage drop.

D.21 G3_selectStandardSection.m

```

1 function [selected_section, ~, standard_sections_data] =
    G3_selectStandardSection(s_required, material)
2 if strcmp(material, 'Copper'), data = [1.5,24,0.5; 2.5,32,0.8;
    4,42,1.2; 6,54,1.7; 10,73,2.8; 16,98,4.5; 25,129,7.0; 35,158,9.5;
    50,188,13.0; 70,232,18.0; 95,275,24.0; 120,315,30.0];
3 else, data = [16,76,3.0; 25,100,4.5; 35,123,6.0; 50,146,8.0;
    70,180,11.5; 95,215,15.0; 120,245,19.0; 150,280,23.0]; end
4 standard_sections_data = struct('section', num2cell(data(:,1)), '
    ampacity', num2cell(data(:,2)), 'cost_per_meter', num2cell(data(:,3))
    );
5 idx = find(data(:,1) >= s_required, 1, 'first');
6 if isempty(idx), selected_section = standard_sections_data(end);
7 else, selected_section = standard_sections_data(idx); end
8 end

```

Listing 21: Selects the next available standard conductor section from a table.

D.22 G4_calculateLifecycleCost.m

```

1 function [total_cost, cable_cost, loss_cost] = G4_calculateLifecycleCost
    (section_data, lineType, d, I, sigma, years, hours_per_year,
    cost_per_kWh)
2 s = section_data.section; cost_per_meter = section_data.cost_per_meter;
3 if strcmp(lineType, 'three-phase'), num_conductors = 3; else,
    num_conductors = 2; end
4 cable_cost = d * cost_per_meter * num_conductors;
5 total_resistance = (1/sigma) * d / s;
6 power_loss_watts = num_conductors * (I^2) * total_resistance;
7 total_kWh_lost = (power_loss_watts / 1000) * hours_per_year * years;
8 loss_cost = total_kWh_lost * cost_per_kWh;
9 total_cost = cable_cost + loss_cost;
10 end

```

Listing 22: Calculates the lifecycle cost (initial + losses) for a given section.

D.23 G5_plotEconomicAnalysis.m

```

1 function [] = G5_plotEconomicAnalysis(sections, cable_costs, loss_costs,
    total_costs, s_technical, s_optimal)
2 figure;
3 section_labels = arrayfun(@(x) num2str(x.section), sections, '
    UniformOutput', false);
4 x_categories = categorical(section_labels);

```

```

5 x_categories = reordercats(x_categories, section_labels);
6 bar_data = [cable_costs', loss_costs'];
7 bar(x_categories, bar_data, 'stacked');
8 hold on;
9 plot(x_categories, total_costs, 'd-r', 'LineWidth', 2, 'MarkerFaceColor',
    , 'r', 'MarkerSize', 8, 'DisplayName', 'Total Lifecycle Cost');
10 optimal_idx = find([sections.section] == s_optimal);
11 if ~isempty(optimal_idx)
12     text(x_categories(optimal_idx), total_costs(optimal_idx), '    Optimal
    ', 'Color', 'red', 'FontWeight', 'bold');
13 end
14 technical_idx = find([sections.section] == s_technical);
15 if ~isempty(technical_idx)
16     text(x_categories(technical_idx), total_costs(technical_idx), '
    Technical Min.', 'Color', 'black', 'VerticalAlignment', 'top');
17 end
18 title('Economic Analysis of Conductor Sizing');
19 xlabel('Standard Conductor Section (mm^2)');
20 ylabel('Total Lifecycle Cost (EUR)');
21 legend({'Initial Cable Cost', 'Cost of Energy Losses', 'Total Lifecycle
    Cost'}, 'Location', 'northoutside', 'Orientation', 'horizontal');
22 grid on; hold off;
23 end

```

Listing 23: Plots the economic analysis results to find the optimal section.

D.24 G6_verifyFinalDesign.m

```

1 function final_vd_percent = G6_verifyFinalDesign(lineType, d, I, cos_phi
    , sigma, s_final, U_source)
2 if strcmp(lineType, 'three-phase'), phase_factor = sqrt(3); else,
    phase_factor = 2; end
3 deltaU_volts = (phase_factor * d * I * cos_phi) / (sigma * s_final);
4 final_vd_percent = (deltaU_volts / U_source) * 100;
5 end

```

Listing 24: Performs a final verification of the voltage drop for the chosen design.